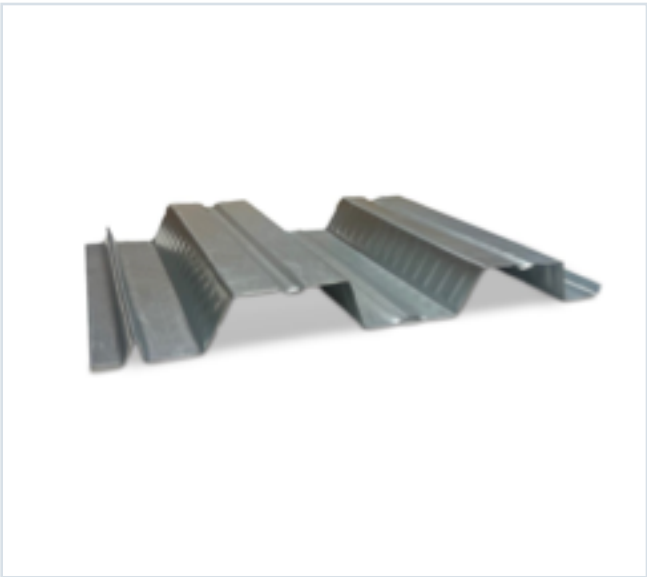


DECKING PRODUCT SUBMITTAL

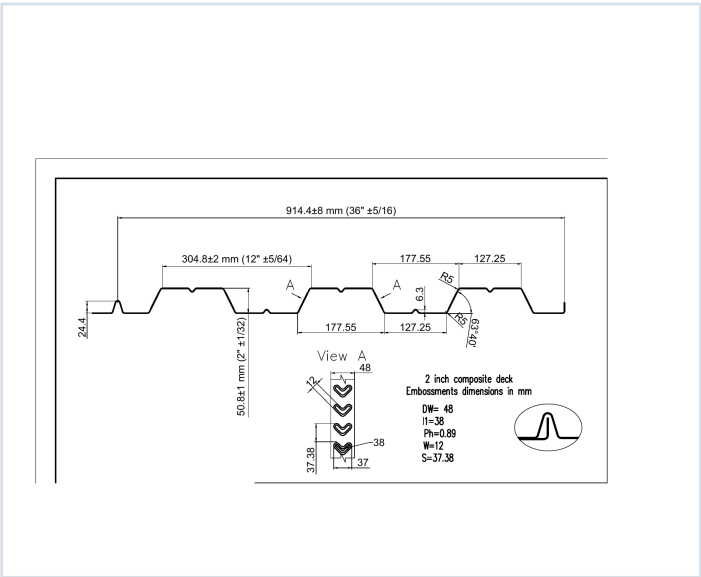
2CD-FY50-16-G40

PRODUCT	GRADE	GAUGE	COATING
2" Composite Deck	Grade 50	16 ga	G40

PRODUCT VIEW



PROFILE DRAWING



SECTION PROPERTIES · 16 GAUGE

Thickness (in.)	Weight (psf)	Fy (ksi)	S+ (in ³ /ft)	S- (in ³ /ft)	M+ (in-lb/ft)	M- (in-lb/ft)	I+ (in ⁴ /ft)	I- (in ⁴ /ft)
0.0598	3.2	50	0.645	0.644	19321	19271	0.703	0.693

SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 54, 55, 56, 57, 58, 59, 60, 61
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

PUBLISHED PERFORMANCE DATA

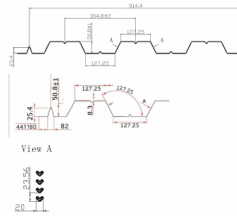
2CD-FY50-16-G40 · 16 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 54

GRADE 50 STEEL



Section Properties

Gage	Design Thickness (inches)	Weight (pcf)	E _x (ksi)	S _x (in ³)	S _y (in ³)	Se (in ⁴)	ASD (E _x + 16E _y)	I _x + (in ⁴)	I _y + (in ⁴)
22	0.0205	16	50	0.247	0.254	2427	2000	0.31	0.290
20	0.0358	19	50	0.328	0.336	3860	3000	0.393	0.373
18	0.0474	21	50	0.408	0.500	14700	14900	0.548	0.530
16	0.0598	23	50	0.465	0.644	19100	19200	0.703	0.683

Note: All section properties and ASD flexural strengths are calculated in accordance with ANSI/AISC 360-2017, AISI S100-2012 and AISI S100-2016.

Shear and Web Crippling

Gage	V _u (kips)	Web Crippling (R _v /3), kips		Web Crippling (R _v /3), kips	
		End Bearing	Interior Bearing	End Bearing	Interior Bearing
22	388	434	476	529	627
20	278	295	327	369	434
18	346	366	403	445	510
16	440	460	507	559	634

Note: All section properties and ASD flexural strengths are calculated in accordance with ANSI/AISC 360-2017, AISI S100-2012 and AISI S100-2016.

STEEL CATALOG PAGE 55

Span	Gage	6'	7'	8'	9'	10'	11'	12'	13'	14'	15'	16'
Single	22	137	133	127	121	115	109	103	97	91	85	79
	20	183	174	165	156	147	138	129	120	111	102	93
	18	273	258	244	230	216	202	188	174	160	146	132
	16	358	338	319	299	279	259	239	219	199	179	159
Double	22	131	125	119	113	107	101	95	89	83	77	71
	20	186	177	168	159	150	141	132	123	114	105	96
	18	277	264	251	238	225	212	199	186	173	160	147
	16	357	337	317	297	277	257	237	217	197	177	157
Triple	22	176	169	162	155	148	141	134	127	120	113	106
	20	243	233	223	213	203	193	183	173	163	153	143
	18	346	324	302	280	258	236	214	192	170	148	126
	16	446	418	390	362	334	306	278	250	222	194	166

Notes:
1. All section properties and ASD (E_x + 16E_y) uniform loads are calculated in accordance with ANSI/AISC 360-2017, AISI S100-2012 and AISI S100-2016.
2. Loads shown in tables are uniformly distributed superimposed loads in psi. Span lengths include center-to-center spacing of supports. Tabulated loads shall not be increased by assuming clear span dimensions.
3. Bending Moment formula used for flexural stress limitations are: Single and Two Span: $M = \frac{wL^2}{8}$; Three Span: $M = \frac{wL^2}{10}$.

Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

Uniform Superimposed Service Load that Causes L/240 Deflection (PSF)

Span	Gage	6'	7'	8'	9'	10'	11'	12'	13'	14'	15'	16'
Single	22	13	13	12	11	10	9	8	7	6	5	4
	20	18	17	16	15	14	13	12	11	10	9	8
	18	27	26	25	24	23	22	21	20	19	18	17
	16	36	35	34	33	32	31	30	29	28	27	26
Double	22	13	13	12	11	10	9	8	7	6	5	4
	20	18	17	16	15	14	13	12	11	10	9	8
	18	27	26	25	24	23	22	21	20	19	18	17
	16	36	35	34	33	32	31	30	29	28	27	26
Triple	22	13	13	12	11	10	9	8	7	6	5	4
	20	18	17	16	15	14	13	12	11	10	9	8
	18	27	26	25	24	23	22	21	20	19	18	17
	16	36	35	34	33	32	31	30	29	28	27	26

Note: For loads that cause L/120 Deflection, multiply by 2.0; For loads that cause L/180 Deflection, multiply by 1.5; For loads that cause L/360 Deflection, multiply by 0.667.

PUBLISHED PERFORMANCE DATA

2CD-FY50-16-G40 · 16 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 56

Normal Weight Concrete (145 pcf)				Lightweight Concrete (115 pcf)			
Maximum Unshored Clear Span				Maximum Unshored Clear Span			
400 (#200) 39 PSF	2x12x22 ga	7'-8"	9'-1"	9'-3"	2x12x22 ga	8'-6"	9'-9"
	2x12x20 ga	9'-3"	10'-6"	10'-10"	2x12x20 ga	10'-0"	11'-3"
	2x12x18 ga	10'-5"	12'-9"	12'-5"	2x12x18 ga	11'-2"	13'-9"
	2x12x16 ga	11'-4"	14'-2"	13'-4"	2x12x16 ga	12'-0"	14'-8"
450 (#250) 45 PSF	2x12x22 ga	7'-7"	8'-8"	8'-10"	2x12x22 ga	8'-3"	9'-5"
	2x12x20 ga	8'-10"	10'-0"	10'-4"	2x12x20 ga	9'-7"	10'-10"
	2x12x18 ga	9'-11"	12'-2"	12'-0"	2x12x18 ga	10'-9"	13'-3"
	2x12x16 ga	10'-11"	13'-8"	12'-10"	2x12x16 ga	11'-8"	14'-6"
500 (#300) 51 PSF	2x12x22 ga	7'-3"	8'-4"	8'-6"	2x12x22 ga	7'-8"	9'-1"
	2x12x20 ga	8'-6"	9'-7"	9'-11"	2x12x20 ga	9'-3"	10'-6"
	2x12x18 ga	9'-7"	11'-8"	11'-8"	2x12x18 ga	10'-5"	12'-9"
	2x12x16 ga	10'-6"	13'-3"	12'-6"	2x12x16 ga	11'-4"	14'-2"
550 (#350) 57 PSF	2x12x22 ga	6'-11"	8'-0"	8'-2"	2x12x22 ga	7'-7"	8'-9"
	2x12x20 ga	8'-3"	9'-3"	9'-6"	2x12x20 ga	8'-11"	10'-1"
	2x12x18 ga	9'-3"	11'-3"	11'-4"	2x12x18 ga	10'-0"	12'-4"
	2x12x16 ga	10'-1"	12'-9"	12'-2"	2x12x16 ga	11'-0"	13'-11"
600 (#400) 63 PSF	2x12x22 ga	6'-9"	7'-9"	7'-10"	2x12x22 ga	7'-5"	8'-6"
	2x12x20 ga	7'-8"	8'-11"	9'-2"	2x12x20 ga	8'-8"	9'-9"
	2x12x18 ga	8'-11"	10'-10"	11'-0"	2x12x18 ga	9'-9"	11'-10"
	2x12x16 ga	9'-9"	12'-4"	11'-10"	2x12x16 ga	10'-8"	13'-6"
650 (#450) 69 PSF	2x12x22 ga	6'-8"	7'-6"	7'-7"	2x12x22 ga	7'-2"	8'-3"
	2x12x20 ga	7'-9"	8'-7"	8'-10"	2x12x20 ga	8'-5"	9'-5"
	2x12x18 ga	8'-8"	10'-6"	10'-9"	2x12x18 ga	9'-5"	11'-6"
	2x12x16 ga	9'-6"	11'-8"	11'-7"	2x12x16 ga	10'-4"	12'-4"

Note
With crippling and clear span not shown accounted for in these tables. Required bearing should be determined based on specific span conditions.

STEEL CATALOG PAGE 61

Slab Thickness (Inches)	Weight (pcf)	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
4	31	400	400	375	330	290	260	230
4.5	35	400	400	400	400	395	397	384
5	39	400	400	400	400	400	370	337
5.5	44	400	400	400	400	400	400	393
6	48	400	400	400	400	400	400	400
6.5	53	400	400	400	400	400	400	400

Slab Thickness (Inches)	10'-6"	11'-0"	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"
4	290	280	272	257	243	230	220	210
4.5	296	298	290	280	276	260	247	236
5	304	295	290	277	268	255	240	230
5.5	304	300	291	285	272	255	240	230
6	307	307	304	304	298	285	270	260
6.5	307	307	307	304	304	298	285	274

16 ga Lightweight Concrete (115 pcf, F_c = 3,000 psi)

Slab Thickness (Inches)	Weight (pcf)	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
4	31	400	400	400	400	392	323	290
4.5	35	400	400	400	400	400	390	360
5	39	400	400	400	400	400	400	400
5.5	44	400	400	400	400	400	400	400
6	48	400	400	400	400	400	400	400
6.5	53	400	400	400	400	400	400	400

Slab Thickness (Inches)	10'-6"	11'-0"	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"
4	260	256	250	236	229	214	200	190
4.5	267	267	261	246	238	220	204	190
5	277	277	270	254	246	228	214	200
5.5	286	286	279	262	254	236	222	208
6	296	296	289	272	264	246	232	218
6.5	306	306	300	282	274	256	242	228